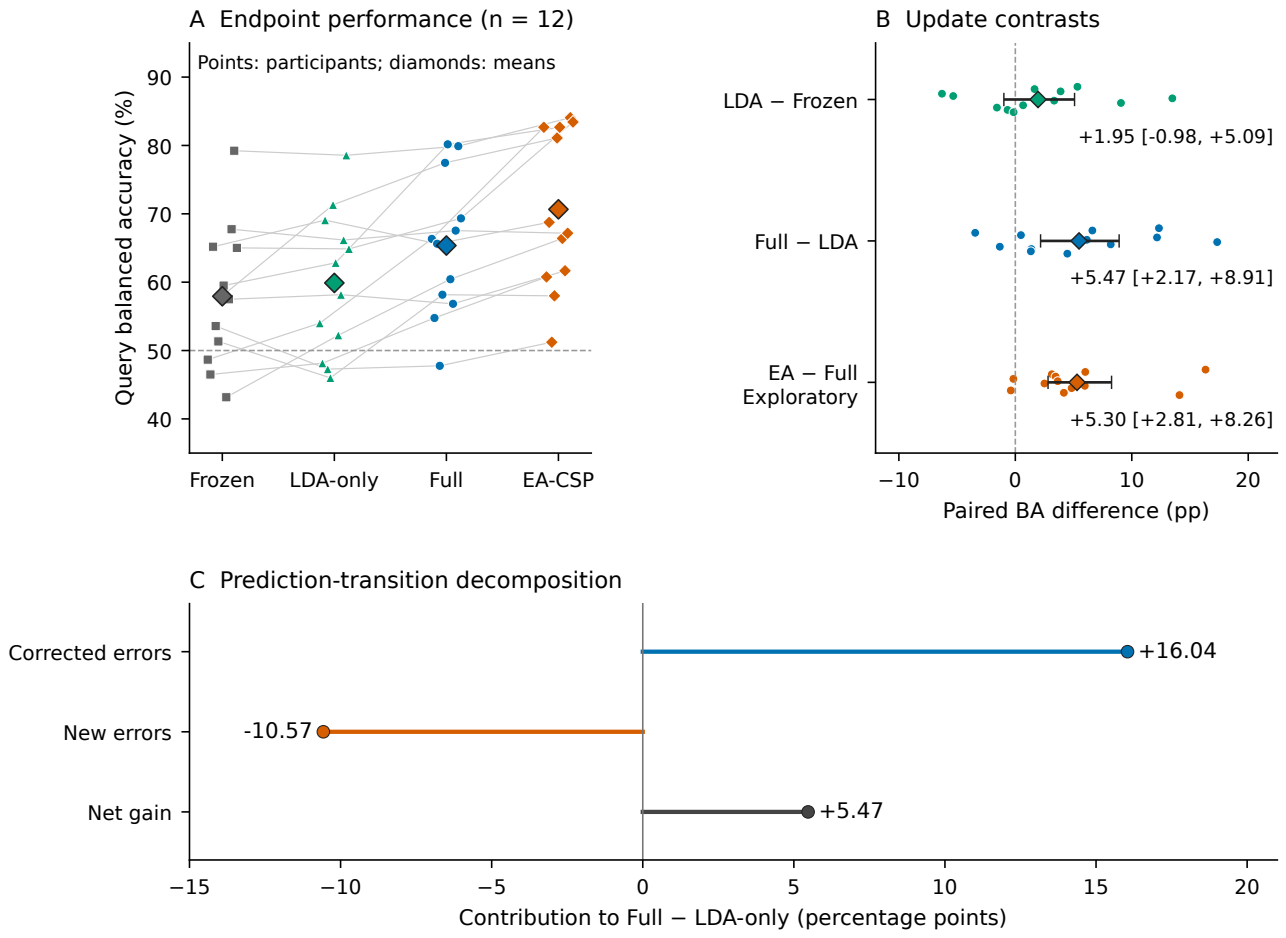
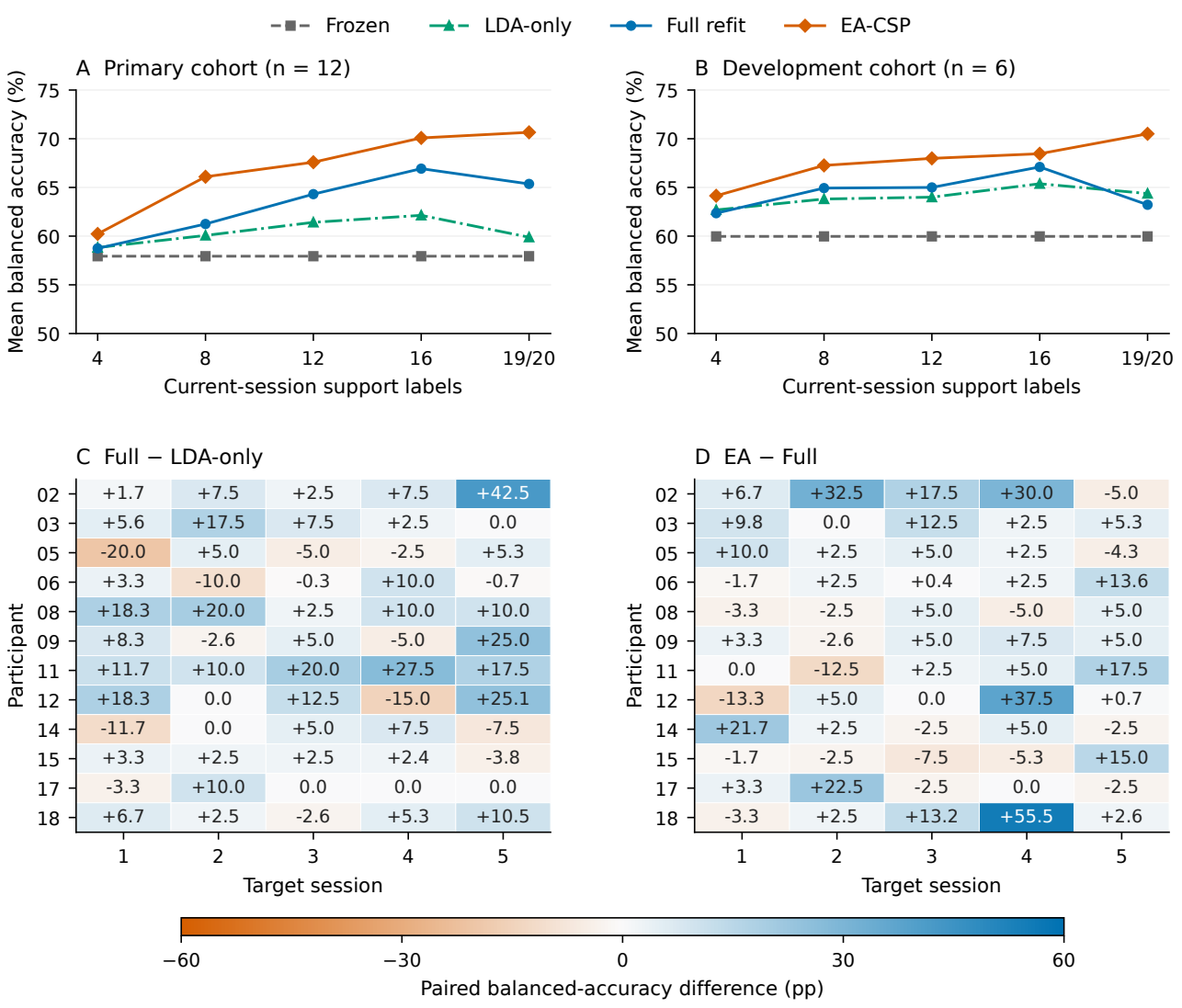
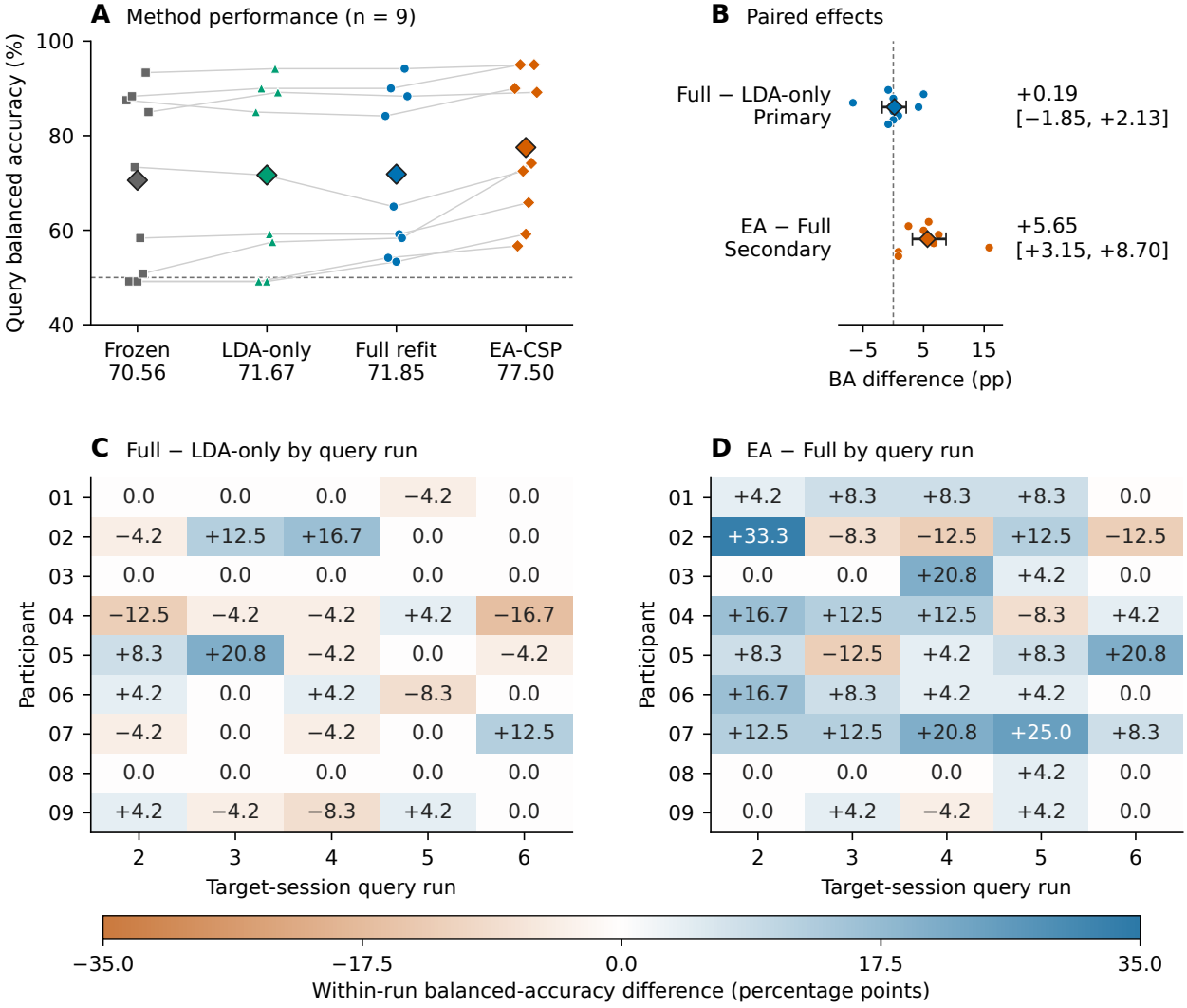
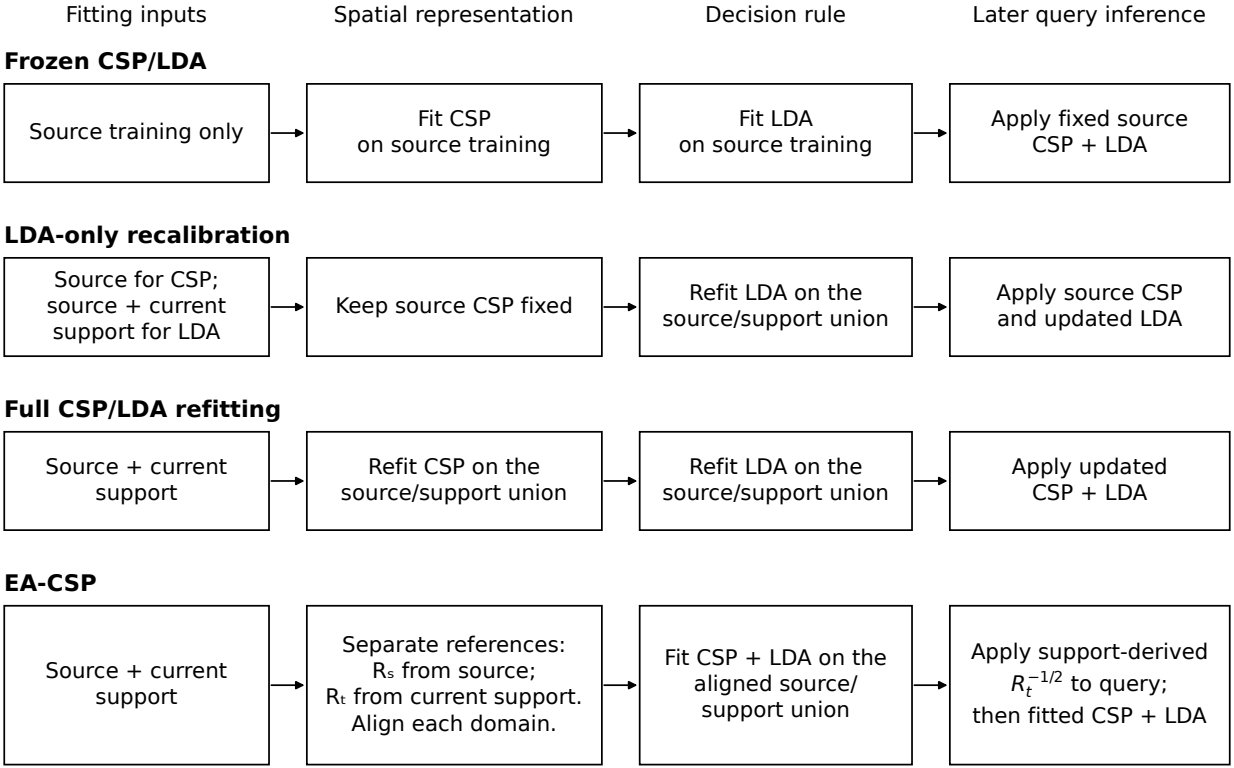

A Endpoint performance (n = 12)
Points: participants; diamonds: means
Query balanced accuracy (%)
90
80
70
60
50
40
Frozen
LDA-only
Full
EA-CSP
B Update contrasts
LDA − Frozen
+1.95 [-0.98, +5.09]
Full − LDA
+5.47 [+2.17, +8.91]
EA − Full Exploratory
+5.30 [+2.81, +8.26]
−10
0
10
20
Paired BA difference (pp)
C Prediction-transition decomposition
Corrected errors
+16.04
New errors
-10.57
Net gain
+5.47
−15
−10
−5
0
5
10
15
20
Contribution to Full − LDA-only (percentage points)


Frozen
LDA-only
Full refit
EA-CSP
A Primary cohort (n = 12)
B Development cohort (n = 6)
Mean balanced accuracy (%)
75
70
65
60
55
50
4
8
12
16
19/20
Current-session support labels
C Full − LDA-only
D EA − Full
Participant
02 +1.7 +7.5 +2.5 +7.5 +42.5
03 +5.6 +17.5 +7.5 +2.5 0.0
05 -20.0 +5.0 -5.0 -2.5 +5.3
06 +3.3 -10.0 -0.3 +10.0 -0.7
08 +18.3 +20.0 +2.5 +10.0 +10.0
09 +8.3 -2.6 +5.0 -5.0 +25.0
11 +11.7 +10.0 +20.0 +27.5 +17.5
12 +18.3 0.0 +12.5 -15.0 +25.1
14 -11.7 0.0 +5.0 +7.5 -7.5
15 +3.3 +2.5 +2.5 +2.4 -3.8
17 -3.3 +10.0 0.0 0.0 0.0
18 +6.7 +2.5 -2.6 +5.3 +10.5
02 +6.7 +32.5 +17.5 +30.0 -5.0
03 +9.8 0.0 +12.5 +2.5 +5.3
05 +10.0 +2.5 +5.0 +2.5 -4.3
06 -1.7 +2.5 +0.4 +2.5 +13.6
08 -3.3 -2.5 +5.0 -5.0 +5.0
09 +3.3 -2.6 +5.0 +7.5 +5.0
11 0.0 -12.5 +2.5 +5.0 +17.5
12 -13.3 +5.0 0.0 +37.5 +0.7
14 +21.7 +2.5 -2.5 +5.0 -2.5
15 -1.7 -2.5 -7.5 -5.3 +15.0
17 +3.3 +22.5 -2.5 0.0 -2.5
18 -3.3 +2.5 +13.2 +55.5 +2.6
1
2
3
4
5
Target session
−60
−30
0
30
60
Paired balanced-accuracy difference (pp)


A Method performance (n = 9)
Query balanced accuracy (%)
100
80
60
40
Frozen
70.56
LDA-only
71.67
Full refit
71.85
EA-CSP
77.50
B Paired effects
Full − LDA-only
Primary
+0.19
[−1.85, +2.13]
EA − Full
Secondary
+5.65
[+3.15, +8.70]
−5
5
15
BA difference (pp)
C Full − LDA-only by query run
Participant
01 0.0 0.0 0.0 −4.2 0.0
02 −4.2 +12.5 +16.7 0.0 0.0
03 0.0 0.0 0.0 0.0 0.0
04 −12.5 −4.2 −4.2 +4.2 −16.7
05 +8.3 +20.8 −4.2 0.0 −4.2
06 +4.2 0.0 +4.2 −8.3 0.0
07 −4.2 0.0 −4.2 0.0 +12.5
08 0.0 0.0 0.0 0.0 0.0
09 +4.2 −4.2 −8.3 +4.2 0.0
2 3 4 5 6
Target-session query run
D EA − Full by query run
Participant
01 +4.2 +8.3 +8.3 +8.3 0.0
02 +33.3 −8.3 −12.5 +12.5 −12.5
03 0.0 0.0 +20.8 +4.2 0.0
04 +16.7 +12.5 +12.5 −8.3 +4.2
05 +8.3 −12.5 +4.2 +8.3 +20.8
06 +16.7 +8.3 +4.2 +4.2 0.0
07 +12.5 +12.5 +20.8 +25.0 +8.3
08 0.0 0.0 0.0 +4.2 0.0
09 0.0 +4.2 −4.2 +4.2 0.0
2 3 4 5 6
Target-session query run
−35.0
−17.5
0.0
17.5
35.0
Within-run balanced-accuracy difference (percentage points)



Source labels: Kumar 40; BNCI 96. Current support labels: Kumar 19/20; BNCI 20.
Frozen uses zero target labels; the three updates share source/support identities.
Query is used for inference, not reference estimation, parameter fitting or selection.
Dataset-specific preprocessing and covariance estimators are defined in Methods.